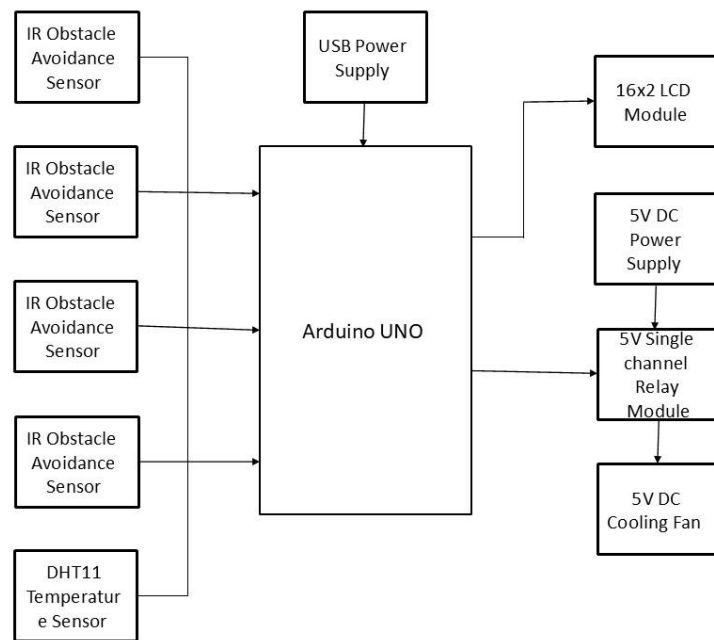


1. Block diagram



2. PROBLEM STATEMENT

Students often spend a lot of time looking for an empty seat in the library, this wastes study time and can cause frustrations especially during exam times. The temperature inside the library can also become uncomfortable, making it difficult for students to concentrate.

3. DESCRIPTION OF THE PROJECT

This project is an Arduino -based smart library system. The system uses four infrared obstacle avoidance sensors to detect whether each seat is occupied or available. A DHT11 Temperature sensor measures the temperature inside the library model. The Arduino UNO processes the information from the sensors and displays the available seats in the 16x2 LCD display. When the temperature is higher than the set value the Arduino activates the Relay to switch on a 5V dc cooling fan and when the temperature drops, the fan switches off automatically.

4. SECTIONS I INTEND TO DESIGN ON MY OWN

I intend to design and build the following parts of the project myself

- Draw the block diagram
- Build the library model using Cardboards and paper chairs
- Connect all electronic components on the breadboard
- Program the Arduino Uno using the Arduino IDE
- Test each sensor to make sure it works correctly
- Connect the relay and cooling fan
- Test the complete system and make any necessary improvements